



OPEN Research on GNNs with stable learning

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Traditional Graph Neural Network (GNN) learning patterns can only achieve optimal performance under the assumption of independent and identically distributed data. However, in real-world scenarios, numerous unpredictable factors give rise to the Out-of-Distribution (OOD) problem. Moreover, this distribution discrepancy may lead to unreliable predictions by GNNs models within unknown OOD domains. Therefore, this paper proposes a machine learning method to enhance the stability of GNN cross-site classification from the perspective of stable learning. The aim is to extract genuine causal features while eliminating spurious causal features. By introducing a feature sample weighting decorrelation technique in the random Fourier transform space and combining it with a baseline GNN model, a Stable-GNN model and a constrained sampling weight gradient update algorithm are designed. Its theoretical proof indicates that this algorithm can ensure the decrease in the loss, thus remedying the shortcomings of the original algorithm in weight updating. While guaranteeing predictive performance on the training distribution data, it also reduces prediction bias on data from unseen test distributions. Compared with the GNN model and the Stable-GNN model (S-GNN), the experimental results of the S-GNN model demonstrate that it not only surpasses the current state-of-the-art GNN models, but also offers a flexible framework for strengthening existing GNNs.

Keywords Graph neural networks, Stable learning, Out-of-distribution, Feature sample weighting decorrelation technique

Graph neural networks (GNNs)¹ are successful deep learning models that extend from unstructured data learning to structured data learning in recent decades. It has been applied to various aspects^{1–4}, among which the main prediction task is the entire graph prediction task, such as molecular graph property prediction^{5–8}, scene graph classification^{9–12}, social network category classification^{7,13–15}, etc. GNNs capture subtle hidden representations in the data by aggregating information from neighboring nodes to extract graph feature information.

Learning patterns for GNNs learn the parameters of GNNs from the training set topology map and perform classification prediction on the unknown test set topology map. This learning mode can achieve the best performance under the assumption of Independent and Identically Distributed (i.i.d.), that is, the training and testing maps are independent of the identically distributed sampling¹⁶. However, there are many unpredictable factors in real life. Such as data selection bias, confounding factors, or other particularities^{17–20}, etc., that is, the training data set and the test data set are not in the same distribution, which is called OOD shift²¹. This is an important reason why most GNNs models fail to make stable predictions. As described in Open Graph Benchmark (OGB)⁵, the GNNs model will suffer 5.66–20% degradation when segmenting graph datasets with different structures according to OOD Settings.

Considering the performance of GNN models on unknown domains, we consider a more realistic and challenging setting where the domain of the testing data is unknown and the latent domain is assumed to be balanced. With this goal, a series of studies on causal learning have been proposed^{22–25}. Considering that statistical dependence between relevant and irrelevant features is the main cause of model collapse under distribution shift, they proposed to remove spurious correlations between dummy features by de-correlation of relevant and irrelevant features to achieve distributional generalization. Since there is no additional supervision to separate relevant features from irrelevant ones, a conservative solution is to de-correlate all features. Recently, this concept has been shown to be effective in improving the generalizability of linear models. In paper²⁶, a sample weighting method is proposed to decorrelate the input variables, and paper²⁷ theoretically proves why this sample weighting can make the linear model produce stable predictions under distribution shift. But they were all developed within the constraints of a linear framework. To deal with more complex data types such as images, Xingxuan Zhang et al.²⁸ proposed a nonlinear feature decorrelation method based on Random Fourier Features (RFF)²⁹ with linear computational complexity.

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Recently, researchers have proposed a series of studies on stable learning^{30–34} to solve the problem of OOD degradation of machine learning methods. Stable learning obtains predictive stability by transforming the training distribution into a stable distribution, which reduces the prediction bias on data from the unseen test distribution while ensuring predictive performance on data from the training distribution. Research^{23,26} proposed the first two learning algorithms, Deep Global Balancing Regression (DGBR) and the Sample Reweighted Decorrelation Operator (SRDO), whose core idea is to obtain the independence between features through weighted sampling techniques to improve the stability of prediction. Recently, stable networks that combine stable learning with CNN have been proposed and have achieved good results in image recognition tasks²⁸. The comparison results also show that the robust network does a good job of removing spurious interference features, such as background, in the training image distribution while preserving task-relevant causal features. From the perspective of causal discovery, it is pointed out that stable learning has the ability of causal feature discovery, which is the main reason to ensure the stability of cross-site prediction²⁸.

In order to address the problems of poor stability and weak generalization ability of the graph network, this paper constructs a GNN architecture based on stable learning based on the literature²⁸, optimizing the intervention technology of weighted sampling of samples, studying the combination method of intervention technology and GNNs and the training algorithm based on gradient descent, trying to eliminate false causal features in the whole training process of the neural network. The generalization of the model and the stability of cross-site automatic diagnosis are improved. Experimental results show that stable-GNN model achieves the best performance on single-site, and cross-site data set classification prediction.

Related work

In this section, we review some important concepts closely related to our work.

GNNs

GNNs are a class of neural network models that are used to deal with graph-structured data^{1,4,13}. Unlike traditional neural networks that deal with regular data structures, GNNs specialize in dealing with data with irregular graph structures⁴. GNNs form new node representations by aggregating the features of each node with the features of its surrounding nodes¹³.

The structure of GNNs usually consists of multiple layers, each of which contains operations such as node embedding^{35,36}, message passing^{37,38}, and pooling^{39,40}. In the node embedding operation, the features of each node are transformed into a low-dimensional vector representation for the neural network to learn and process⁴¹. In a message-passing operation, each node receives information from its neighbors and aggregates this information into a new node representation⁴². In the pooling operation, the node representations are merged into the whole graph representation to facilitate the prediction of graph-level tasks⁴⁰. Many publications show that traditional GNN models achieve remarkable results in i.i.d. environments¹⁶.

Stable learning

Stable learning⁴³ is a method to reduce the variance of machine learning in the unknown test set under the premise of ensuring its prediction performance. In stable learning, stable distribution refers specifically to underlying data distribution properties that maintain model performance as data distribution changes. The fundamental idea behind stable learning is to identify the key characteristics of various categories, eliminate irrelevant features and erroneous correlations, and then base predictions solely on the key characteristics⁴³. Since correlations between features can affect the predictions of a model, several studies^{26,28,44} have been devoted to removing such correlations during training. Some pioneering studies based on the Lasso framework^{45,46} proposed to remove relevant characteristics by adding a rule, which stipulates that highly correlated features cannot be selected simultaneously.

Random fourier features

RFF²⁹ is an efficient approximation technique based on kernel methods that significantly improves the computational efficiency of large-scale machine learning problems by explicitly mapping nonlinear kernel functions to low-dimensional feature spaces. The method's foundation lies in the application of Bochner's theorem, which facilitates the transformation of the implicit mapping in the regenerative kernel Hilbert space (RKHS) into an explicit D-dimensional Euclidean space representation through the use of stochastic Fourier features. This transformation leads to a significant reduction in the computational complexity of the traditional kernel method, shifting it from $O(n^2)$ to $O(nD)$, where n signifies the sample size and D denotes the dimensionality of the stochastic features, which are considerably smaller than n . Cui et al.²⁸ innovatively combine the RFF technique with the Frobenius paradigm to construct an efficient independence testing framework. The method computes the Frobenius paradigm of the partial cross-covariance matrix as a statistic, and when the value of the paradigm converges to zero, it can be determined that the variables satisfy asymptotic independence. This innovation not only inherits the computational efficiency advantage ($O(nD)$ complexity) of the RFF method, but also provides a practical tool with theoretical guarantee for correlation analysis of high-dimensional data.

Learning sample weights for decorrelation

The core principle of Learning Sample Weights for Decorrelation (LSWD) lies in deriving instance-specific weights that, when applied to the training data, effectively suppress undesirable correlations (e.g., spurious associations between features and target variables). Sample weighting assigns a weight to each training instance to modulate its importance in the loss function. In conventional machine learning, all samples are typically assigned uniform weights (i.e., $\omega_i = 1$ for all i , indicating equal importance). The objective of LSWD is to learn an optimal weighting scheme that minimizes specified spurious correlations in the reweighted data distribution.

Dataset	Properties					
	Number of graphs	Number of classes/targets	Number of average nodes	Number of average edges	Vertex labels	Edge labels
ENZYMES	600	6	32.6	62.14	+	–
PROTEINS_full	1113	2	39.06	72.82	+	–
NCI109	4127	2	29.68	32.13	+	–
DD	1178	2	284.32	715.66	+	–
SYNTHETICnew	300	2	100.00	196.25	–	–
MUTAG	188	2	17.93	19.79	+	+
Mutagenicity	4337	2	30.32	30.77	+	+
COIL-DEL	3900	100	21.54	54.24	–	+

Table 1. TUDataset statistics and properties.

Dataset	TaskType	Graphs	Average nodes	Average edges	Task	Metric
Molbase	Binary classification	1513	34.1	36.9	1	ROC-AUC
Molbbbp	Binary classification	2039	24.1	26.0	1	ROC-AUC
Molhiv	Binary classification	41127	25.5	27.5	1	ROC-AUC
Molclintox	Multi-label classification	1477	26.2	27.9	2	ROC-AUC
Moltox21	Multi-label classification	7831	18.6	19.3	12	ROC-AUC
Molesol	Regression	1128	13.3	13.7	1	RMSE
Mollipo	Multi-label classification	4200	27.0	29.5	1	RMSE

Table 2. Summary of real-world datasets.

Investigate⁴³ According to theoretical analysis, making sure that the input variables are independent aids in determining which important variables have the greatest impact on the prediction. The following is the analysis. Take into account the following methods of data generation, such as selection bias and confounding: Using $[Y = f(S) + \varepsilon]$, where $[X = (S, V)]$ and $Y \perp V | S$, Y is the outcome variable, S is the input variable corresponding to the direct cause of Y , V is another input variable that may have spurious correlation with Y . If $f(x)$ is a nonlinear function and a linear model such as ordinary least squares (OLS) is used for fitting, due to model misspecification, the coefficient of the irrelevant variable V may be incorrectly estimated as a non-zero value, leading to predictions that depend on spurious associations. Weighted OLS can cause the coefficient of V to be zero after sample weights have been optimized using stable learning techniques like the Sample Reweighting Correlation Operator (SRDO)²⁶. Prediction stability is maintained even in cases when the joint distribution of (S, V) changes (for example, in distribution shift or bias scenarios) since this property guarantees that the model only uses the true variable S for predictions.

Method

Datasets

The data sets we employ are from the TUDataset⁴⁷ and OGB datasets⁵. Table 1 presents the data of TUDataset, and Table 2 presents the data of OGB datasets. TUDataset, a widely used dataset library for graph machine learning that contains datasets covering a variety of domains, including chemistry^{5,48,49}, biology^{50–52}, social networks^{53,54}, etc.

The data sets in the TUDataset are graph-based datasets, where each sample is a graphical object. Each graph consists of a set of nodes (vertices) and edges (edges connecting nodes), along with optional node features and graph labels⁴⁷. Node features are attributes or feature vectors associated with a node, while graph labels are labels or categories of the entire graph.

We tested the GNNs model on some selected datasets from the TUDataset, using a random 10-fold cross-validation (split ratio 70/10/20) on the datasets, and observe the performance of stationary learning on monocentric graph data. In addition, we use the Grad-CAM⁵⁵ technique to interpret MUTAG⁵⁶, a commonly used graph classification data set. It consists of 4337 molecular graphs. Each graph is classified into two categories according to its mutagenic effect.

We used seven molecular property prediction datasets in OGB datasets⁵ to test the performance of the stable graph model in OOD mode. All molecules were pre-processed using RDKit⁵⁷. Each graph represents a molecule where the nodes are atoms and the edges are chemical bonds. We used the 9-dimensional node features and 3D edge features provided by⁵, which have better generalization performance. The task of these data sets is to predict the molecular properties of the target, for example, whether a molecule inhibits HIV viral replication⁵. The input edge features are 3D and contain bond types, bond stereochemistry, and additional bond features indicating whether a bond is conjugate or not.

Depending on the nature of the molecules, these data sets can be divided into three tasks: binary classification, multilabel classification, and regression⁵. Unlike commonly used random splitting, these datasets employ a

scaffold splitting procedure⁵⁸ to split molecules from different scaffolds into training or test sets. All molecules with the same scaffold can be viewed as an environment, and scaffold splitting attempts to separate molecules with different scaffolds into different subsets⁵⁹.

For example, two molecules, cyclopropanol (C_3H_6O) and 1, 4-cyclohexanediol ($C_6H_{12}O_{22}$), contain different scaffold patterns: the former scaffold is 3 ring and the latter is 6C ring. Although the sampling distribution is different, they are all easily soluble in water because their invariant hydroxyl (–OH) subgraph group is attached to different scaffolds⁶⁰. The environments of the training and test sets are different, resulting in different distributions of the graphs. Therefore, this type of data can be used to test the generalizability of the stable GCN model for graphs with different distributions.

Weighted sampling module

The core concept of stable learning lies in the formation of a stable distribution through weighted sampling. The weighted sampling module used in this paper is made up of RFF extractor and a LSWD optimizer^{26,29}.

The RFF extractor uses the random Fourier transform to transform the feature sequence $O \in R^{l \times n}$ output by the last graph convolutional layer of the GCN model into the RFF sequence $\{H_i^j = (h_{i1}^j \dots h_{in}^j) \in R^{1 \times n}\}_{1 \leq j \leq r}$, and then inputs the $\{H_i^j\}_{1 \leq j \leq r}$ into the LSWD optimizer to complete the feature decorrelation in the random Fourier transform space and find the optimal sample weights that match it.

Finally, the statistical independence between two different features O_i and O_k is maximized. Due to that, the independence in the random Fourier transform space is equivalent to the statistical independence in the original space²⁸.

Stochastic Fourier transforms and feature independence

A common approach to non-linearly map data into a higher dimensional space is the inner product kernel map. Usually, we cannot directly obtain the explicit representation of the mapping in high-dimensional space. For instance, the Gaussian kernel function corresponds to infinite-dimensional space. This space is obviously not available. The random Fourier transform, on the other hand, is a low-order method of approximating the kernel function.

The purpose is to approximate the kernel function by d different parameters of the random Fourier transform function $z(\cdot)$: direct mapping of the raw data of D dimensional into a random feature space of d dimensional, where the inner product of two features is approximately equal to the value of the kernel function:

$$z(x)z(y) \approx k(x, y) = k(x - y) \quad (1)$$

In²⁸, Eq. (1) indicates that the statistical independence between the features x and y in the original feature space can be estimated by computing the covariance matrices of the transformed representations $z(x)$ and $z(y)$ in the Reproducing kernel Hilbert space (RKHS)⁶¹. At this point, if the covariance matrices of $z(x)$ and $z(y)$ are zero matrices, then the features x and y are statistically independent of each other.

As mentioned above, we can use the random Fourier transform to approximate the expressions $z(x)$ and $z(y)$ in the RKHS space, and thus transform the covariance matrix in the RKHS space to approximate the computation in the low-dimensional random Fourier transform space.

Definition 1 The random Fourier transform function is given as follows:

$$H_{RFF} = \{h : x \rightarrow \sqrt{2} \cos(\omega x + \phi) \mid \omega \sim N(0, 1), \phi \sim Uniform(0, 2\pi)\} \quad (2)$$

where, ω and ϕ are random frequencies and random phases sampled from the standard normal distribution $N(0, 1)$ and uniform distribution $Uniform(0, 2\pi)$, respectively.

Let there be two one-dimensional random distributions D_x, D_y , and we obtain two n dimensional random sequences $X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_n\}$ by randomly sampling n times on each of the two distributions.

By randomly sampling k frequencies and k phases, respectively, using a random Fourier transform function, we can further transform the sequences X and Y in the original space into $\{H_{x_1}^{\omega, \phi}, \dots, H_{x_n}^{\omega, \phi}\}$ and $\{H_{y_1}^{\omega, \phi}, \dots, H_{y_n}^{\omega, \phi}\}$. Where $H_{x_n}^{\omega, \phi} = (H^{\omega_1, \phi_1}(x_n), \dots, H^{\omega_k, \phi_k}(x_n))$, $H^{\omega_k, \phi_k}(x_n) = \sqrt{2} \cos(\omega_k x_n + \phi_k)$.

At this point, the estimated covariance matrix Σ_{xy} with respect to $z(x)$ and $z(y)$ in the random Fourier transform space can be written as follows:

$$\Sigma_{xy} = \frac{1}{n-1} \sum_{i=1}^n \left(H_{x_i}^{\omega, \phi} - \frac{1}{n} \sum_{j=1}^n H_{x_j}^{\omega, \phi} \right)^T \times \left(H_{y_i}^{\omega, \phi} - \frac{1}{n} \sum_{j=1}^n H_{y_j}^{\omega, \phi} \right) \quad (3)$$

The statistical independence measure I_{xy} of Y and X is defined as the Frobenius norm value⁶² of the matrix of covariance moment Σ_{xy} , that is, $I_{xy} = \|\Sigma_{xy}\|_F^2$. When I_{xy} decreases to 0, X and Y gradually tend to be independent of each other.

Feature decorrelation

In this section, feature correlations based on the spatial independence criterion of the RFF above $I_{xy} = 0$ using a sample weighting method are not taken into account. After sample weighting, the weighted covariance matrix of the random feature sequence X, Y in Eq. (3) can be re-expressed as follows:

$$\Sigma_{xy}(W) = \frac{1}{n-1} \sum_{i=1}^n \left(w_i H_{x_i}^{\omega, \phi} - \frac{1}{n} \sum_{j=1}^n w_j H_{x_j}^{\omega, \phi} \right)^T \times \left(w_i H_{y_i}^{\omega, \phi} - \frac{1}{n} \sum_{j=1}^n w_j H_{y_j}^{\omega, \phi} \right) \quad (4)$$

Here, $W = (w_1, \dots, w_n) \in R_+^n$ denotes the sample weight vector, which satisfies $\sum_{j=1}^n w_j = n$ and $w_i \geq 0$.

For m different features $X^1, \dots, X^m$, we maximize the independence between pairwise features by minimizing the Frobenius norm value of the weighted population covariance. The specific optimization formula is as follows:

$$\begin{aligned} J(W) = \min \sum_{i,j=1, i \neq j}^m \|\Sigma_{X^i X^j}(W)\|_F^2 \\ \text{s.t.} \begin{cases} \sum_{j=1}^n w_j = n; \\ w_i \geq 0. \end{cases} \end{aligned} \quad (5)$$

Gradient update algorithm for sampling weights

In the actual optimization process, we optimize the weighted classification loss and the loss of feature independence of the Eq. (5) in rotation. Suppose that the m features output by feature extractor $f^{(t)}$ and predictor $g^{(t)}$ obtained by us in the t round of iterative optimization are respectively $X^1(t), \dots, X^m(t)$, then the rotation optimization is as follows:

$$\begin{aligned} W^{(t)} = \min_{\sum_{j=1}^n w_j = n, w_i \geq 0} \sum_{i,j=1, i \neq j}^m \|\Sigma_{X^i(t) X^j(t)}(W^{(t-1)})\|_F^2 \\ f^{(t+1)}, g^{(t+1)} = \min_{f,g} \sum_{i=1}^n w_i^{(t)} L(g^{(t+1)}(X^i(t)), y_i) \end{aligned} \quad (6)$$

The optimization Eq. (6) needs to be satisfied in the constraint $\sum_{j=1}^n w_j^{(t)} = n, w_i^{(t)} \geq 0$. The usual gradient descent algorithm cannot satisfy the constraints. In this paper, a gradient update algorithm is specially designed to satisfy this constraint, as follows:

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- 1: **Initial setup** $w_i = 1, 1 \leq i \leq n$, Learning rate: $\lambda > 0$
 - 2: **Find the loss gradient** $\Delta w = \frac{dl}{dw}$ of Eq. (7)
 - 3: **Use zero equalization** $\Delta w = \Delta w - m(\Delta w), m(\Delta w) = \frac{1}{n} \sum_{i=1}^n w_i$ **to update** $w = w - \lambda \Delta w$
 - 4: **Nonnegative** $w \geq 0$. If $w_i < 0$, reset $w_i = 0$.
 - 5: **Normalized** $w, \bar{w} = \frac{n}{\sum_{i=1}^n w_i} \times w$
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Algorithm 1. Gradient update algorithm for sampling weights

The above algorithm can be used to ensure that the loss of Eq. (5) decreases when w after each update satisfies the constraint Δ_n , and the specific proof is as follows:

- **Step 1: Initial setup** $w_i = 1, 1 \leq i \leq n$, **Learning rate:** $\lambda > 0$.
- **Step 2: Find the loss gradient** $\Delta w = \frac{dl}{dw}$ of Eq. (7) The convergence of the classical gradient descent algorithm is based on the first-order Taylor series approximation:

$$L(w) = L(w(0)) + \left(\frac{dL}{dw(0)} \right)^T (w - w(0)) \quad (7)$$

Where $w(0)$ is the weight vector obtained in the previous round and w is the weight vector that must be determined by the gradient descent algorithm in this round. According to the above equation, the direction of the negative gradient is the direction of descent, that is, $w - w(0) = -\frac{dL}{dw(0)}$, and the use of $w = w(0) - \lambda \frac{dL}{dw(0)}$ update ensures that the loss decreases. The changes we make are in steps 3, 4 and 5.

- **Step 3: Use zero equalization** $\Delta w = \Delta w - m(\Delta w)$, $m(\Delta w) = \frac{1}{n} \sum_{i=1}^n w_i$ to update $w = w - \lambda \Delta w$ Updating w using zero equalize $\Delta w(0)$ can be written as follows:

$$w = w(0) - \lambda \left(\frac{dl}{dw(0)} - m(\Delta w(0)) e \right) \quad (8)$$

where e is a n dimensional all-ones vector. Furthermore, it can be computed from Eq. (8)

$$\begin{aligned} L(w) - L(w(0)) &= -\lambda \left(\frac{dl}{dw(0)} \right)^T \left(\frac{dl}{dw(0)} - m(\Delta w(0)) e \right) \\ &= -\lambda \left(\left(\frac{dl}{dw(0)} \right)^T \frac{dl}{dw(0)} - m(\Delta w(0)) \left(\frac{dl}{dw(0)} \right)^T e \right) \\ &= -\lambda \left(\left(\frac{dl}{dw(0)} \right)^T \frac{dl}{dw(0)} - nm^2(\Delta w(0)) \right) \end{aligned} \quad (9)$$

Accordingly, let vector $\frac{dl}{dw(0)} = (s_1, \dots, s_n)^T$, we have:

$$\begin{aligned} m(\Delta w(0)) &= \frac{1}{n} \sum_{i=1}^n s_i \\ \left(\frac{dl}{dw(0)} \right)^T \frac{dl}{dw(0)} &= \sum_{i=1}^n s_i^2 \end{aligned} \quad (10)$$

Subsequently, from the basic inequality, the following is obtained:

$$\sum_{i=1}^n s_i^2 \geq \frac{1}{n} \left(\sum_{i=1}^n s_i \right)^2 = nm^2(\Delta w(0)) \quad (11)$$

Here, Eq. (11) shows that when $n \geq 2$, there must be: $\left(\frac{dl}{dw(0)} \right)^T \frac{dl}{dw(0)} \geq nm^2(\Delta w(0))$. By further analyzing Eq. (9), $L(w) - L(w(0)) < 0$ can be obtained. Therefore, we conclude that updating w with zero equalization $\Delta w(0)$ also ensures a decrease in the loss function as in Eq. (5).

- **Step 4: Nonnegative** $w \geq 0$. If $w_i < 0$, reset $w_i = 0$. Vector $\frac{dl}{dw(0)} = (s_1, \dots, s_n)^T$, mean $m(\Delta w(0)) = b$, so $\lambda \left(\frac{dl}{dw(0)} - m(\Delta w(0)) e \right) = \lambda(s_1 - b, \dots, s_n - b)^T$. Now, let us prove $\sum_{i=1}^n s_i(s_i - b) \geq 0$.

Proof Let us write w_i as $w_i = w_i(0) - \lambda(s_i - b)$. If $w_i < 0$, there must be $s_i - b > 0$. According to the non-negation operation, we need to decrease $s_i - b$ to t_i in the range of ≥ 0 .

At this time, if $s_i \geq 0$, due to $t_i \geq 0$, the modified $s_1(s_1 - b) + \dots + s_i t_i + \dots + s_n(s_n - b)$ is still ≥ 0 , which can ensure the loss of Eq. (5) decreases.

If $s_i < 0$, then $s_i - b > t_i \geq 0$, $s_i(s_i - b) < s_i t_i < 0$ can be obtained. This means that $s_1(s_1 - b) + \dots + s_i t_i + \dots + s_n(s_n - b)$ is still ≥ 0 , ensuring that Eq. (5) loss decreases. Thus, we conclude that a further non-negative w after a zero mean $\Delta w(0)$ still ensures a decrease in Eq. (5) loss. $\square$

- **Step 5: Normalized** w . $\bar{w} = \frac{\sum_{i=1}^n w_i}{\sum_{i=1}^n w_i} \times w$ Let $w^T(0) e = n$, we first easily see that using the zero-averaged gradient to update $w(0)$, the updated w must also satisfy $w^T e = n$. Then non-negative w . In this case, we will increase the w_i from less than 0 to 0, which means that w must have $w^T e > n$ after non-negative. Finally, we normalize w so that it satisfies $\alpha w^T e = n$ by scaling down, clearly scaling factor $\alpha < 1$. Substituting it into Eqs. (4), (5) and (6), it can be seen that when using $\alpha < 1$ to scale w , it also scales the matrix $\Sigma_{X^i X^j}(W)$, so it will make $L(w)$ further decrease. The complexity of computing the sample weights is $O(B * N)$, where B is the epoch and N is the feature dimension. Certificate completed.

Stable-GNN

Finding the correct causality between feature and result can improve the stability and generalization of network⁴³. GNN can only find the correlation between the sample features, but not the causality. In order to find a causal relationship between features and results, we need to take some measures to make the features as independent as possible between samples.

The overall architecture of Stable-GNN is shown in Fig. 1, Stable-GNN eliminates linear and nonlinear dependencies between features by utilizing the characteristics of RFF and sample weighting²⁶. The feature map is drawn through GNN and then entered the classifier to calculate the predicted loss when the dependence is removed by the weighting processing of the sample. The final loss is obtained by combining the weight of the processed sample with the predicted loss.

Learning sample weights globally

Global learning of sample weights in deep learning tasks is frequently plagued by issues of storage space and computing expense. When utilising the gradient optimisation approach, only a subset of the samples in each batch may be observed, preventing the global weights of all samples from being learned. In response to this difficulty, this study uses the weight saving and reloading technique introduced by Peng Cui²⁸ et al. to optimise sample weights by combining features and sample weights gained during the training phase as global knowledge. In particular, assuming batch size B , a join operation along the sample dimension is used to combine the batch features and weights into a globally optimised feature Z and weight W :

$$W = \text{Concat}(\omega_1, \omega_2, \dots, \omega_k, \omega_l), Z = \text{Concat}(z_1, z_2, \dots, z_k, z_l)$$

$z_1, z_2, \dots, z_k$ and $\omega_1, \omega_2, \dots, \omega_k$, respectively, stand for the global features and weights that are dynamically modified during each training batch. This technique lowers the compute and storage overhead from $O(N)$ to $O(kB)$.

Results

Experimental setup: The TUDataset is commonly applied in the field of graph machine learning and has achieved relatively good performance on many network models. In order to verify the effectiveness of our Stable Graph Network model architecture in removing spurious causal relationships on the basis of graph networks, we only compare it with the baseline models.

For datasets with edge features of graph connection, such as MUTAG, Mutagenicity, and COIL-DEL, the baseline networks used are GCN (Fig. 2), GINE⁶³, GINE0⁶³, and ⁶⁴. For data sets without edge features of graph connection, the baseline networks used are GCN, GIN, and GIN0. The number of layers for the GCN model is set to 3, and the main GCN parameters in the experiment are: the number of graph convolution layer channels (search range [24, 48, 96, 192]), the initial learning rate is 0.001, and the Adam optimizer is used.

To reduce the risk of overfitting, a dropout strategy with a fixed parameter of 0.5 is used in the second and third layers of the graph convolution network. The number of layers for the GIN, GIN0, GINE, and GINE0 networks is set to 5, and the main parameters of these models are: the number of channels of the graph convolution layer (search range [8, 16, 32, 64, 128]), the learning rate is selected from 0.1, 0.2, ..., 1, and the SGD optimizer is used.

For the Stable Graph Network, we set 20 iterations to train the sample weights, with the weight learning rate taken from the range [0.1, 0.2, ..., 1], and 300 iterations to train the entire model. For all methods, the batch sample size is searched in the range [8, 16, 32, 64], and the model with the best validation performance is used for prediction.

Case 1: Experimental results on the TUDataset. The experimental results shown in Tables 3, 4 indicate that in datasets without edge features of the graph connection, only the Stable-GCN model increased the average accuracy compared to the GCN model, while the Stable-GIN0 model and the Stable-GIN model had a lower average accuracy compared to the GIN0 and GIN models, but the overall difference is not large. In datasets with graph connection edge features, the average accuracy of the Stable-GCN model and the Stable-GINE

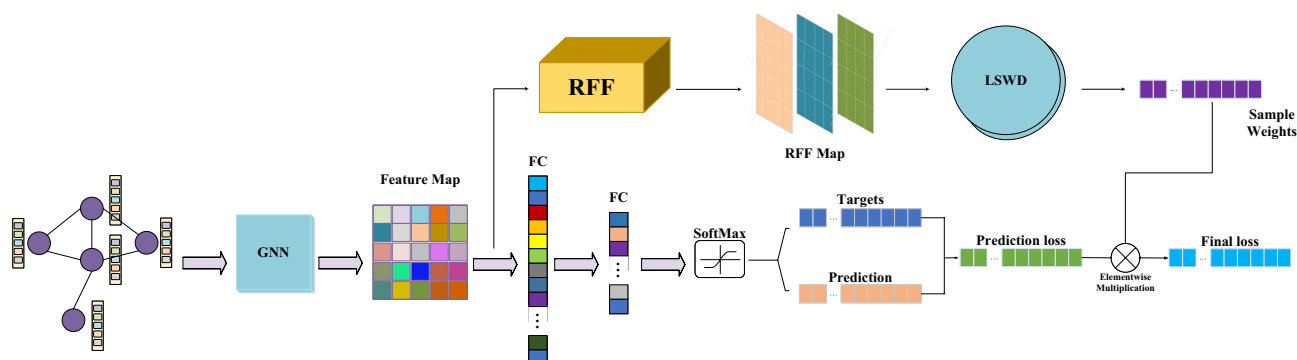


Figure 1. Framework for stable GNNs.

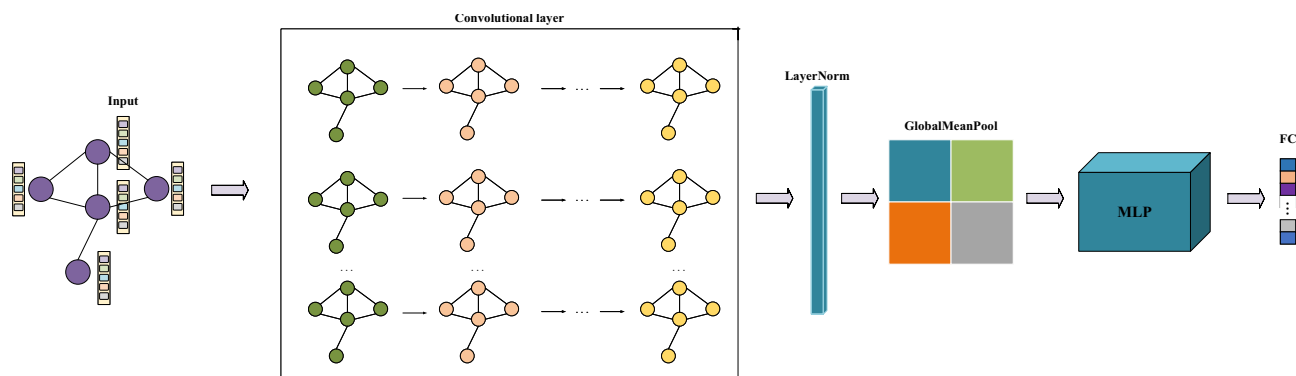


Figure 2. Framework for graph convolutional neural networks.

	GCN	Stable-GCN	GIN0	Stable-GIN0	GIN	Stable-GIN
ENZYMES	0.3967±0.0770	0.4150±0.0337	0.3233±0.0761	0.3667±0.0645	0.4033±0.0909	0.4067±0.0523
PROTEINS_full	0.7340±0.0397	0.7430±0.0440	0.7386±0.0408	0.7368±0.0390	0.7413±0.0269	0.7332±0.0425
NCI109	0.7259±0.0255	0.7238±0.0189	0.7543±0.0211	0.7451±0.0129	0.7504±0.0295	0.7415±0.0188
DD	0.7156±0.0375	0.7190±0.0589	0.7131±0.0489	0.7165±0.0350	0.7097±0.0483	0.6952±0.0421
SYNTHETICnew	0.5867±0.0980	0.6200±0.0686	0.6367±0.0706	0.5800±0.0777	0.5967±0.0547	0.5867±0.0581

Table 3. Results on TUDatasets.

	GCN	Stable-GCN	GINE0	Stable-GINE0	GINE	Stable-GINE
MUTAG	0.8345±0.0971	0.8348±0.0845	0.6594±0.1163	0.7126±0.1323	0.7129±0.1394	0.7184±0.1331
Mutagenicity	0.7971±0.0239	0.8010±0.0201	0.6225±0.0527	0.6698±0.0628	0.6370±0.0594	0.6745±0.0582
COIL-DEL	0.3028±0.0334	0.3128±0.0323	0.3321±0.0298	0.3549±0.0732	0.4333±0.0271	0.4203±0.0545

Table 4. Results on TUDatasets.

model is not much different from the GCN model and the GINE model, respectively. However, the average accuracy of the Stable-GINE0 model is 4.1% *higher* than the original GINE0 models. We believe that the reason for this phenomenon is that the causal associations learned by the stable network are partially different in structure from the key functional networks learned by the original models. Stable learning can effectively remove spurious correlations by learning the weight features of the edges and relying on this clue for prediction. Overall, stable learning has improved the stability performance of graph network models in the classification of single-site datasets.

Case 2: experimental setup on the OGB datasets. In this part, we apply our Stable-GNNs algorithm to seven OGB datasets for OOD graph property prediction.

- **Baselines:** Since the superiority of GNNs over traditional methods in graph-level tasks, such as kernel-based methods^{51,65}, has been demonstrated in previous literature⁷, here we mainly apply two widely-used GNN baseline models, namely GCN² and GIN⁶³.
- For all GNNs baseline models, we follow⁵ and set the number of graph network layers to 5. For binary classification and multilabel classification datasets, the dropout value is set to 0.5, while for regression datasets, the dropout value is set to 0.0. The batch size for binary and multilabel classification as well as regression datasets is selected from [16, 32, 64, 128]. The hidden layer dimension is 300. The learning rate is selected from [0.1, 0.2, ..., 1] using the Adam optimizer. For the stable graph networks, we set 20 iterations to train the sample weights, with the weight learning rate taken from the range [0.1, 0.2, ..., 1], and 100 iterations to train the entire model. The model with the best validation performance is used for prediction.
- These datasets typically use chemical bond types as their edge types, and to include edge features, we follow⁵ and add the transformed edge features to the input node features.
- For the OGB datasets, we use the metrics recommended in the original papers⁵ for each task.
- The experimental results on the OGB datasets, as shown in Table 5, indicate that in the classification tasks, the Stable-GIN network and the Stable-GCN network achieved the first and second rankings respectively, with a clear improvement in classification performance over the baseline GNNs models. The best-performing Stable-GIN model achieved a 3.72% higher AUC value on the molbase dataset compared to the GIN model.

Methods	Binary classification			Multilable classification		Average ROC-AUC	Regression		
	Molhiv ROC-AUC	Molbase ROC-AUC	Molbbbp ROC-AUC	Molclintox ROC-AUC	Moltox21 ROC-AUC		Molesol RMSE	Mollipo RMSE	Average RMSE
GCN	0.7557	0.7593	0.6908	0.8706	0.7034	0.75596	1.1118	0.7696	0.9407
Stable-GCN	0.7684	0.7639	0.6936	0.9049	0.7261	0.77138	0.9123	0.7578	0.83505
GIN	0.7707	0.75569	0.6881	0.88036	0.7066	0.76029	0.9827	0.7142	0.84845
Stable-GIN	0.782	0.7929	0.7197	0.9016	0.7155	0.78234	0.9648	0.6655	0.81515

Table 5. Results on OGB datasets.

- In the regression tasks, the average RMSE values of the Stable-GIN and Stable-GCN models are lower than the baseline GNNs models, and the average RMSE value of Stable-GCN is 0.1056 lower than the GCN model, demonstrating that our models have improved the stability of the baseline GNNs models and their effectiveness in the OOD generalization problem. The superior performance of Stable-GCN/GIN on datasets with three different tasks and a wide range of data scales shows that the proposed framework is sufficiently generic and applicable to datasets with various properties.

Experiments on the effect of weighted sampling on model stability

To verify the effect of increased sample independence on model stability, we designed ablation experiments to compare model performance under different correlation retention ratios. Specifically, we set six correlation levels: 0%, 20%, 40%, 60%, 80%, and 100%. The neural network models we employ are all GIN network models. Among them:

- 0% correlation ($WX = 0$) indicates that the samples are completely independent;
- 20% correlation ($0.2X$) indicates that 20% of the original correlation is retained;
- other percentages ($0.4X, 0.6X, 0.8X, X$) and so on, where X indicates that 100% of the original correlation is retained.

The results of the experiment are displayed in Fig. 3. In the categorization task, experimental results demonstrate a decreasing trend in the ROC-AUC values of the model as the sample correlation (WX) increases. The most significant decreasing trend is observed in subfigure (a) and (e). Furthermore, subfigure (f) demonstrates that the ROC-AUC values decrease in conjunction with decreasing sample independence (i.e., increasing correlation) across all categorical datasets. As demonstrated in subfigure (g), (h), and (i), the root mean square error (RMSE) of the model exhibits an upward trend in conjunction with the reduction of data feature independence (i.e., correlation enhancement). This observation signifies that the model's stability diminishes. The experiments verify the effectiveness of the weighted sampling strategy, which can significantly improve the stability of the model by eliminating spurious correlation features (enhanced independence), and exhibits better generalization performance in both classification and regression tasks.

Conclusion and future work

This paper has investigated the application of stable learning in graph networks. To address the problem that most GNN models cannot make stable predictions in OOD shift environments, this paper has introduced a feature sample weighting technique in the random Fourier transform space and combines it with baseline GNN models to design a Stable-GNN model. It has designed a constrained sampling weight gradient update algorithm and theoretically proved that the algorithm can guarantee loss decrease, making up for the defects of the original algorithm in updating the weights. Extensive experiments have effectively demonstrated the effectiveness, flexibility, and interpretability of Stable-GNNs. OOD detection is critical for ensuring the reliability and security of machine learning models, particularly in domains demanding high generalization and robustness. Key applications include computer vision⁶⁶, object recognition⁶⁷, multimodal data fusion⁶⁸, autonomous driving in wild environments speech recognition⁶⁹, semantic shift detection in NLP(Natural Language Processing)⁷⁰, anomaly identification in medical imaging⁷¹, defect classification in industrial quality inspection⁷², and environmental distribution adaptation in robotics⁷³. These domains share common challenges: complex high-dimensional data distributions, limited labeled data for supervised learning, and inherent heterogeneity in OOD samples. We argue that our approach can similarly enhance stability learning across other OOD-sensitive fields.



Figure 3. Model performance on the OGB classification and regression datasets as sample correlation (WX) changes.

Data availability

The datasets generated and analyzed during the current study are derived from publicly available sources. The TUDatasets is available at <https://chrsmrrs.github.io/datasets/>, and OGB datasets is available at <https://ogb.stanford.edu/>.

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References

- Scarselli, F., Gori, M., Tsoi, A. C., Hagenbuchner, M. & Monfardini, G. The graph neural network model. *IEEE Trans. Neural Netw.* **20**, 61–80 (2008).
- Kipf, T. N. & Welling, M. Semi-supervised classification with graph convolutional networks. arXiv preprint [arXiv:1609.02907](https://arxiv.org/abs/1609.02907) (2016).
- Veličković, P. et al. Graph attention networks. arXiv preprint [arXiv:1710.10903](https://arxiv.org/abs/1710.10903) (2017).
- Hamilton, W., Ying, Z. & Leskovec, J. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems* Vol. 30 (2017).
- Hu, W. et al. Open graph benchmark: Datasets for machine learning on graphs. In *Advances in Neural Information Processing Systems* Vol. 33, 22118–22133 (2020).
- Lee, J. B., Rossi, R. & Kong, X. Graph classification using structural attention. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* 1666–1674 (2018).
- Ying, Z. et al. Hierarchical graph representation learning with differentiable pooling. In *Advances in Neural Information Processing Systems* Vol. 31 (2018).
- Wieder, O. et al. A compact review of molecular property prediction with graph neural networks. *Drug Discov. Today Technol.* **37**, 1–12 (2020).
- Pope, P. E., Kolouri, S., Rostami, M., Martin, C. E. & Hoffmann, H. Explainability methods for graph convolutional neural networks. In *Proceedings of the IEEE/CVF Conference On Computer Vision and Pattern Recognition* 10772–10781 (2019).
- Li, Y., Chen, R., Zhang, Y., Zhang, M. & Chen, L. Multi-label remote sensing image scene classification by combining a convolutional neural network and a graph neural network. *Remote Sens.* **12**, 4003 (2020).
- Sharifzadeh, S., Baharlou, S. M. & Tresp, V. Classification by attention: Scene graph classification with prior knowledge. In *Proceedings of the AAAI Conference on Artificial Intelligence* Vol. 35, 5025–5033 (2021).
- Monninger, T. et al. Scene: Reasoning about traffic scenes using heterogeneous graph neural networks. *IEEE Robot. Autom. Lett.* **8**, 1531–1538 (2023).
- Zhang, M., Cui, Z., Neumann, M. & Chen, Y. An end-to-end deep learning architecture for graph classification. In *Proceedings of the AAAI Conference on Artificial Intelligence* Vol. 32 (2018).

14. Wang, K. et al. Minority-weighted graph neural network for imbalanced node classification in social networks of internet of people. *IEEE Internet Things J.* **10**, 330–340 (2022).
15. Xie, Y., Lv, S., Qian, Y., Wen, C. & Liang, J. Active and semi-supervised graph neural networks for graph classification. *IEEE Trans. Big Data* **8**, 920–932 (2022).
16. Liao, R., Urtasun, R. & Zemel, R. A pac-Bayesian approach to generalization bounds for graph neural networks. arXiv preprint [arXiv:2012.07690](https://arxiv.org/abs/2012.07690) (2020).
17. Bengio, Y. et al. A meta-transfer objective for learning to disentangle causal mechanisms. arXiv preprint [arXiv:1901.10912](https://arxiv.org/abs/1901.10912) (2019).
18. Engstrom, L., Tran, B., Tsipras, D., Schmidt, L. & Madry, A. Exploring the landscape of spatial robustness. In *International Conference on Machine Learning* 1802–1811 (PMLR, 2019).
19. Su, J., Vargas, D. V. & Sakurai, K. One pixel attack for fooling deep neural networks. *IEEE Trans. Evolut. Comput.* **23**, 828–841 (2019).
20. Hendrycks, D. & Dietterich, T. Benchmarking neural network robustness to common corruptions and perturbations. arXiv preprint [arXiv:1903.12261](https://arxiv.org/abs/1903.12261) (2019).
21. Sun, Y. et al. Test-time training with self-supervision for generalization under distribution shifts. In *International Conference on Machine Learning* 9229–9248 (PMLR, 2020).
22. Shen, Z., Cui, P., Kuang, K., Li, B. & Chen, P. On image classification: Correlation vs causality. arXiv preprint [arXiv:1708.06656](https://arxiv.org/abs/1708.06656) (2017).
23. Kuang, K., Cui, P., Athey, S., Xiong, R. & Li, B. Stable prediction across unknown environments. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* 1617–1626 (2018).
24. Gopnik, A. et al. A theory of causal learning in children: Causal maps and Bayes nets. *Psychol. Rev.* **111**, 3 (2004).
25. Schulz, L. E. & Gopnik, A. Causal learning across domains. *Dev. Psychol.* **40**, 162 (2004).
26. Shen, Z., Cui, P., Zhang, T. & Kunag, K. Stable learning via sample reweighting. In *Proceedings of the AAAI Conference on Artificial Intelligence* Vol. 34, 5692–5699 (2020).
27. Kuang, K., Xiong, R., Cui, P., Athey, S. & Li, B. Stable prediction with model misspecification and agnostic distribution shift. In *Proceedings of the AAAI Conference on Artificial Intelligence* Vol. 34, 4485–4492 (2020).
28. Zhang, X. et al. Deep stable learning for out-of-distribution generalization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* 5372–5382 (2021).
29. Rahimi, A. & Recht, B. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems* Vol. 20 (2007).
30. Sun, Y. et al. Test-time training for out-of-distribution generalization. In *The 37th International Conference on Machine Learning* (2019).
31. Krueger, D. et al. Out-of-distribution generalization via risk extrapolation (rex). In *International Conference on Machine Learning* 5815–5826 (PMLR, 2021).
32. Jin, C. Cross-project software defect prediction based on domain adaptation learning and optimization. *Expert Syst. Appl.* **171**, 114637 (2021).
33. Ziser, Y. & Reichart, R. Task refinement learning for improved accuracy and stability of unsupervised domain adaptation. In *proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* 5895–5906 (2019).
34. Fan, S., Wang, X., Shi, C., Cui, P. & Wang, B. Generalizing graph neural networks on out-of-distribution graphs. In *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023).
35. Zhang, C., Song, D., Huang, C., Swami, A. & Chawla, N. V. Heterogeneous graph neural network. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* 793–803 (2019).
36. Chen, Y., Wu, L. & Zaki, M. Iterative deep graph learning for graph neural networks: Better and robust node embeddings. In *Advances in Neural Information Processing Systems* Vol. 33, 19314–19326 (2020).
37. Gilmer, J., Schoenholz, S. S., Riley, P. F., Vinyals, O. & Dahl, G. E. Message passing neural networks. In *Machine Learning Meets Quantum Physics* 199–214 (2020).
38. Feng, J., Chen, Y., Li, F., Sarkar, A. & Zhang, M. How powerful are k-hop message passing graph neural networks. In *Advances in Neural Information Processing Systems* Vol. 35, 4776–4790 (2022).
39. Liu, C. et al. Graph pooling for graph neural networks: Progress, challenges, and opportunities. arXiv preprint [arXiv:2204.07321](https://arxiv.org/abs/2204.07321) (2022).
40. Grattarola, D., Zambon, D., Bianchi, F. M. & Alippi, C. Understanding pooling in graph neural networks. *IEEE Trans. Neural Netw. Learn. Syst.* **35**, 2708–2718 (2022).
41. Zhou, J., Liu, L., Wei, W. & Fan, J. Network representation learning: from preprocessing, feature extraction to node embedding. *ACM Comput. Surv. (CSUR)* **55**, 1–35 (2022).
42. Zhong, Z., Li, C.-T. & Pang, J. Hierarchical message-passing graph neural networks. *Data Min. Knowl. Discov.* **37**, 381–408 (2023).
43. Cui, P. & Athey, S. Stable learning establishes some common ground between causal inference and machine learning. *Nat. Mach. Intell.* **4**, 110–115 (2022).
44. Xu, R., Cui, P., Shen, Z., Zhang, X. & Zhang, T. Why stable learning works? A theory of covariate shift generalization. arXiv preprint [arXiv:2111.02355](https://arxiv.org/abs/2111.02355) (2021).
45. Takada, M., Suzuki, T. & Fujisawa, H. Independently interpretable lasso: A new regularizer for sparse regression with uncorrelated variables. In *International Conference on Artificial Intelligence and Statistics* 454–463 (PMLR, 2018).
46. Chen, S.-B., Ding, C., Luo, B. & Xie, Y. Uncorrelated lasso. In *Proceedings of the AAAI Conference on Artificial Intelligence* Vol. 27, 166–172 (2013).
47. Morris, C. et al. TUDataset: A collection of benchmark datasets for learning with graphs. arXiv preprint [arXiv:2007.08663](https://arxiv.org/abs/2007.08663) (2020).
48. Chen, G. et al. Alchemy: A quantum chemistry dataset for benchmarking AI models. arXiv preprint [arXiv:1906.09427](https://arxiv.org/abs/1906.09427) (2019).
49. Gilmer, J., Schoenholz, S. S., Riley, P. F., Vinyals, O. & Dahl, G. E. Neural message passing for quantum chemistry. In *International Conference on Machine Learning* 1263–1272 (PMLR, 2017).
50. Borgwardt, K. M. et al. Protein function prediction via graph kernels. *Bioinformatics* **21**, i47–i56 (2005).
51. Shervashidze, N., Schweitzer, P., Van Leeuwen, E. J., Mehlhorn, K. & Borgwardt, K. M. Weisfeiler–Lehman graph kernels. *J. Mach. Learn. Res.* **12**, 2539 (2011).
52. Barabasi, A.-L. & Oltvai, Z. N. Network biology: understanding the cell's functional organization. *Nat. Rev. Genet.* **5**, 101–113 (2004).
53. Yanardag, P. & SVN, V. Deep graph kernels. In *Acm sigkdd International Conference on Knowledge Discovery and Data Mining* 1365–1374 (2015).
54. Viswanath, B., Mislove, A., Cha, M. & Gummadi, K. P. On the evolution of user interaction in Facebook. In *Proceedings of the 2nd ACM Workshop on Online Social Networks* 37–42 (2009).
55. Selvaraju, R. R. et al. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision* 618–626 (2017).
56. Debnath, A. K., Lopez de Compadre, R. L., Debnath, G., Shusterman, A. J. & Hansch, C. Structure-activity relationship of mutagenic aromatic and heteroaromatic nitro compounds. Correlation with molecular orbital energies and hydrophobicity. *J. Med. Chem.* **34**, 786–797 (1991).
57. Landrum, G. Rdkit documentation. *Release* **1**, 4 (2013).
58. Wu, Z. et al. Moleculenet: A benchmark for molecular machine learning. *Chem. Sci.* **9**, 513–530 (2018).

59. Corso, G., Stark, H., Jegelka, S., Jaakkola, T. & Barzilay, R. Graph neural networks. *Nat. Rev. Methods Primers* **4**, 17 (2024).
60. Ishida, S., Miyazaki, T., Sugaya, Y. & Omachi, S. Graph neural networks with multiple feature extraction paths for chemical property estimation. *Molecules* **26**, 3125 (2021).
61. Berlinet, A. & Thomas-Agnan, C. *Reproducing kernel Hilbert spaces in probability and statistics* (Springer, Cham, 2011).
62. Gretton, A., Bousquet, O., Smola, A. & Schölkopf, B. Measuring statistical dependence with hilbert-schmidt norms. In *International Conference on Algorithmic Learning Theory* 63–77 (Springer, 2005).
63. Xu, K., Hu, W., Leskovec, J. & Jegelka, S. How powerful are graph neural networks? arXiv preprint [arXiv:1810.00826](https://arxiv.org/abs/1810.00826) (2018).
64. Xu, K. et al. Representation learning on graphs with jumping knowledge networks. In *International Conference on Machine Learning* 5453–5462 (PMLR, 2018).
65. Shervashidze, N., Vishwanathan, S., Petri, T., Mehlhorn, K. & Borgwardt, K. Efficient graphlet kernels for large graph comparison. In *Artificial Intelligence and Statistics* 488–495 (PMLR, 2009).
66. Wei, J., Wang, G. & Zhang, S. Fine-grained medical image out-of-distribution detection through multi-view feature uncertainty and adversarial sample generation. *Pattern Recogn.* **162**, 111401 (2025).
67. Riesenhuber, M. & Poggio, T. Models of object recognition. *Nat. Neurosci.* **3**, 1199–1204 (2000).
68. Wang, Y. Survey on deep multi-modal data analytics: Collaboration, rivalry, and fusion. In *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* Vol. 17, 1–25 (2021).
69. Bukhsh, Z. & Saeed, A. On out-of-distribution detection for audio with deep nearest neighbors. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 1–5 (IEEE, 2023).
70. Montanelli, S. & Periti, F. A survey on contextualised semantic shift detection. arXiv preprint [arXiv:2304.01666](https://arxiv.org/abs/2304.01666) (2023).
71. Tschuchnig, M. E. & Gadermayr, M. Anomaly detection in medical imaging—a mini review. In *International Data Science Conference* 33–38 (Springer, 2021).
72. Abd-Al-Rahman, M. & Mousavi, A. A review and analysis of automatic optical inspection and quality monitoring methods in electronics industry. *IEEE Access* **8**, 183192–183271 (2020).
73. Bredeche, N., Montanier, J.-M., Liu, W. & Winfield, A. F. Environment-driven distributed evolutionary adaptation in a population of autonomous robotic agents. *Math. Comput. Model. Dyn. Syst.* **18**, 101–129 (2012).

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Author contributions

Conceptualization, Y.X. and B.Y.; methodology, Y.X., W.L. and B.Y.; validation, W.W. and Y.X.; writing—original draft, W.W. and Y.X.; writing—review and editing, P.W. and L.P.; funding acquisition, W.L., L.P. and B.Y. All authors reviewed the manuscript.

Declarations

Conflict of interest

The authors declare that they have no conflict of interest.

Additional information

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